

Task 5

Again, do the same thing as in Task 3, but this time let the domain be from 0 to R_o . That means, while Task 3 was for a hollow cylinder, Task 5 is for a solid cylinder.

Use the following boundary conditions:

$$\begin{aligned} \text{At } r = 0, \quad u &= 0, \\ \text{At } r = R_o, \quad \sigma_{rr} &= -p \end{aligned}$$

Do everything non-dimensionally like in Task 3. When presenting your results, make sure that you compare with the analytical solution.

Implementation:

The dimensionless governing differential equation was solved using the Finite Difference Method (FDM) over the domain $0 \leq \tilde{r} \leq 1$. The computational domain was divided into N uniform intervals, producing $N+1$ nodal points. Central difference approximations were used for the first and second derivatives at the interior nodes.

The boundary condition at the center of the cylinder, $\tilde{r} = 0$, was imposed directly by setting the displacement equal to zero. The stress boundary condition at the outer surface, $\tilde{r} = 1$, was first expressed in terms of displacement and then discretized using a backward finite difference approximation for the derivative.

The resulting set of algebraic equations was assembled into a matrix system of the form $Au = b$ and solved in MATLAB using the matrix division operator. The numerical displacement distribution obtained from the finite difference solution was compared with the analytical solution. The displacement profiles and the maximum absolute error were used to evaluate the accuracy of the numerical method.

Calculation:

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$$\frac{d^2 \tilde{u}}{d\tilde{r}^2} + \frac{1}{\tilde{r}} \frac{d\tilde{u}}{d\tilde{r}} - \frac{\tilde{u}}{\tilde{r}^2} = 0$$

general solution $\tilde{u} = A\tilde{r} + \frac{B}{\tilde{r}}$

boundary condition

$$\tilde{u}(0) = 0$$

But $\lim_{\tilde{r} \rightarrow 0} \frac{B}{\tilde{r}} = \infty$

$\therefore B$ must be $= 0$

$$\therefore \tilde{u} = A\tilde{r}$$

now,

$$\tilde{\sigma}_{rr} = \frac{1}{(1+\nu)(1-2\nu)} \left[(1-\nu) \frac{d\tilde{u}}{d\tilde{r}} + \nu \frac{\tilde{u}}{\tilde{r}} \right] = -1$$

$$2 \quad \frac{d\tilde{u}}{d\tilde{r}} = A \quad \& \quad \frac{\tilde{u}}{\tilde{r}} = A$$

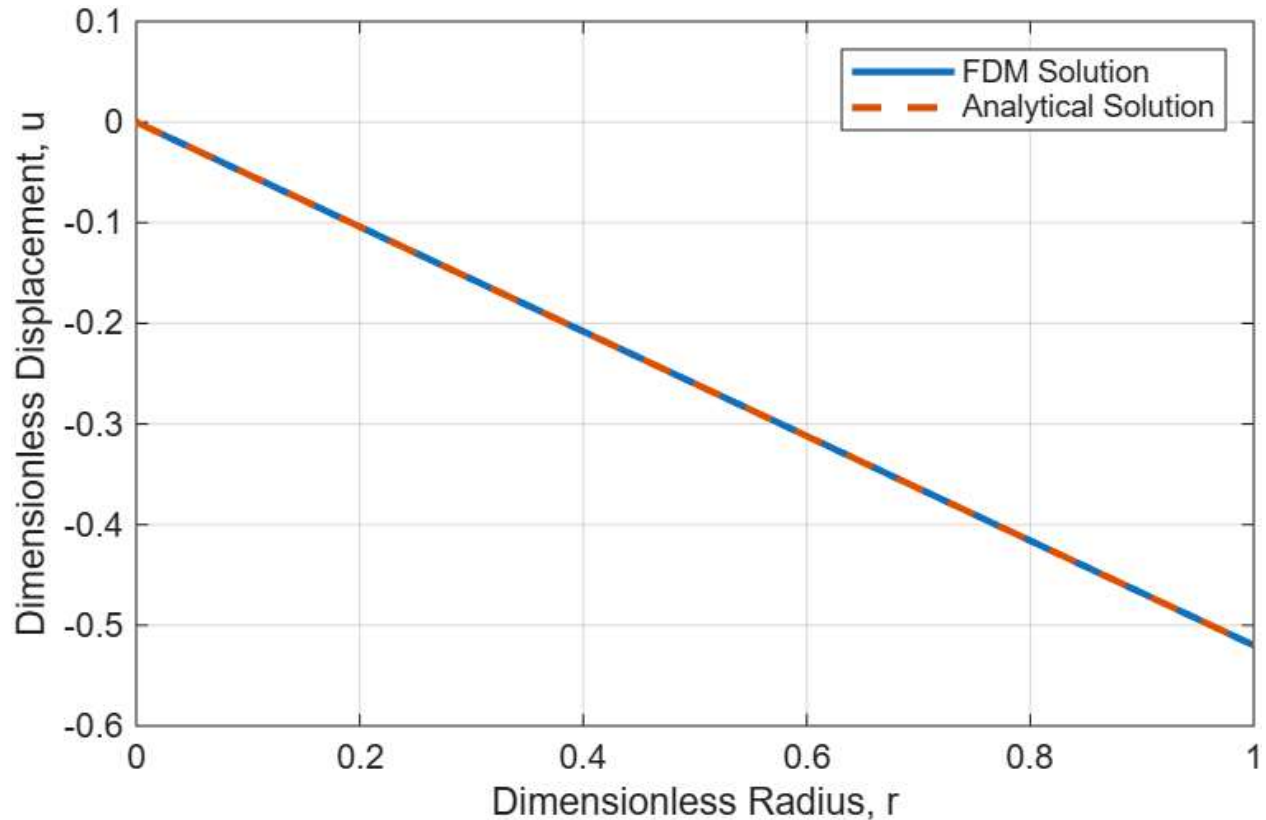
$$\begin{aligned} \therefore \tilde{\sigma}_{rr} &= \frac{((1-\nu) + \nu) A}{(1+\nu)(1-2\nu)} \\ &= \frac{A}{(1+\nu)(1-2\nu)} \end{aligned}$$

FDM follows the same method as Task 3.

Matlab Implementation:

```
clear;
clc;
nu = 0.3;
N = 100;
h = 1/N;
r = linspace(0,1,N+1)';
A = zeros(N+1,N+1);
b = zeros(N+1,1);
A(1,1) = 1;
b(1) = 0;
for j = 2:N
    rj = r(j);
    a = 1/h^2 - 1/(2*h*rj);
    c = -2/h^2 - 1/rj^2;
    d = 1/h^2 + 1/(2*h*rj);
    A(j,j-1) = a;
    A(j,j) = c;
    A(j,j+1) = d;
end
A(N+1,N) = -(1-nu)/h;
A(N+1,N+1) = (1-nu)/h + nu;
b(N+1) = -(1+nu)*(1-2*nu);
u_num = A\b;
u_exact = -(1+nu)*(1-2*nu)*r;
max_error = max(abs(u_num-u_exact));
fprintf('Maximum error = %e\n',max_error); %[output:98eb222e]
figure; %[output:0f276f8d]
plot(r,u_num,'LineWidth',2); %[output:0f276f8d]
hold on; %[output:0f276f8d]
plot(r,u_exact,'--','LineWidth',2); %[output:0f276f8d]
xlabel('Dimensionless Radius, r'); %[output:0f276f8d]
ylabel('Dimensionless Displacement, u'); %[output:0f276f8d]
legend('FDM Solution','Analytical Solution'); %[output:0f276f8d]
grid on; %[output:0f276f8d]
```

Result:



Conclusion: The displacement field of a solid cylinder subjected to external pressure was successfully obtained using the Finite Difference Method. The governing differential equation was discretized over the dimensionless domain and solved numerically in MATLAB. Comparison with the analytical solution showed excellent agreement, indicating that the numerical implementation was accurate. The results demonstrate that the Finite Difference Method provides an effective and reliable approach for solving boundary value problems arising in solid mechanics.

Github Repository: <https://github.com/mugdhajha/Project-Research/>